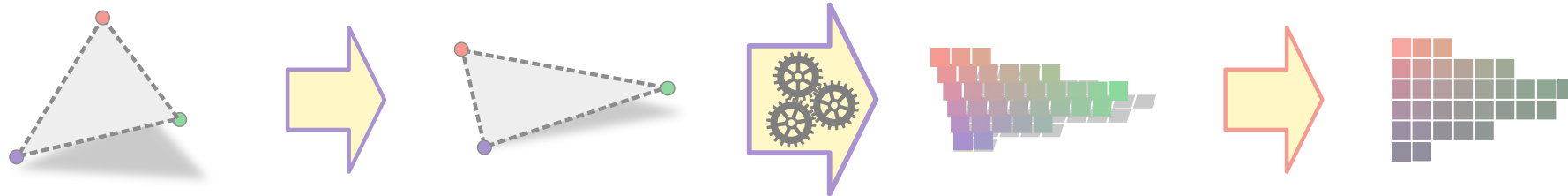


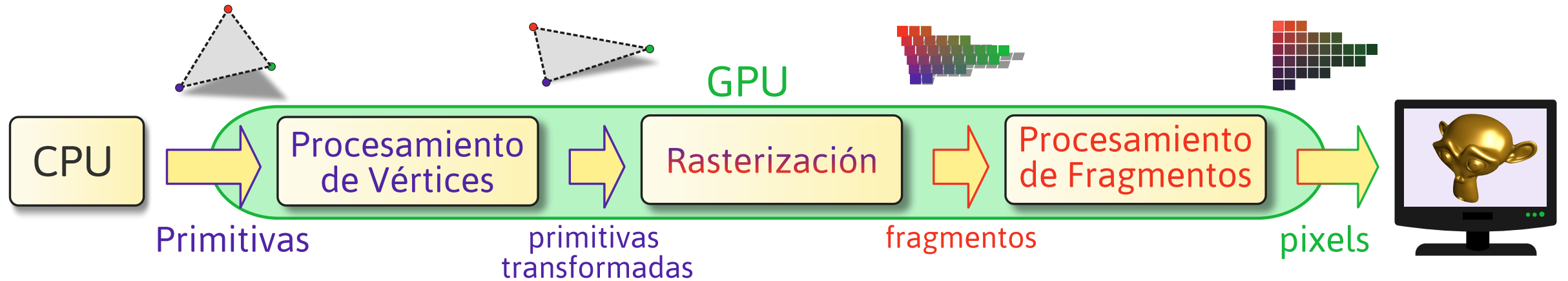
Computación Gráfica 2024

Unidad 2

Rasterización



LA RASTERIZACIÓN EN EL PIPELINE



- ▶ Entrada del pipeline

- ▶ Por cada objeto:

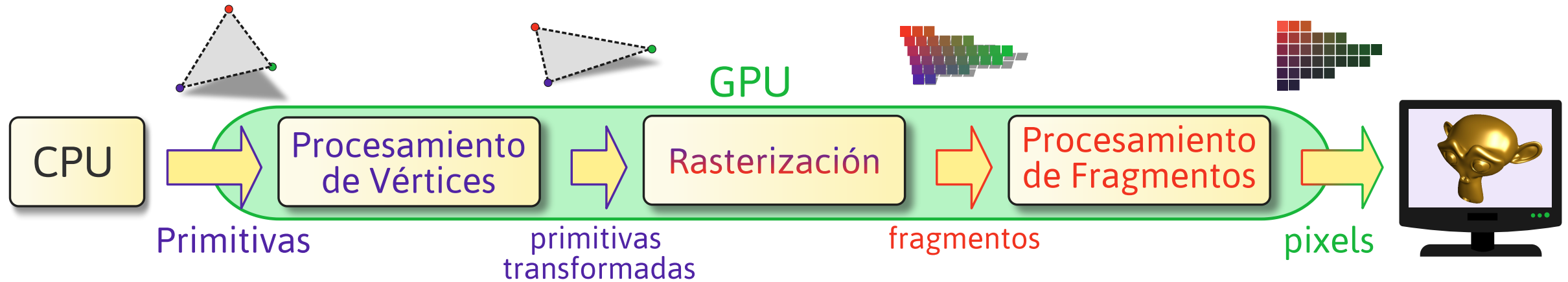
- ▶ Una **mall**a: vértices y conectividades

! *"Curvas y Superficies"*

- ▶ Su "ubicación" en el mundo: matriz "model"

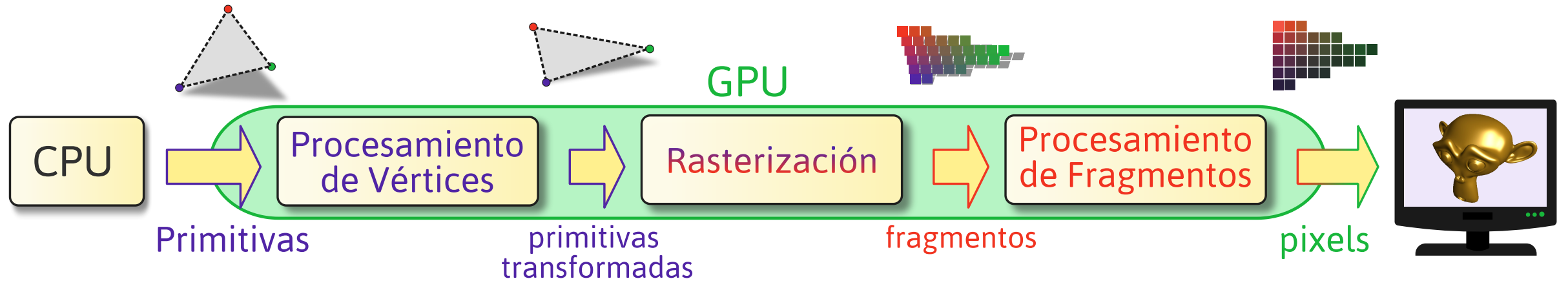
- ▶ Su "material"

LA RASTERIZACIÓN EN EL PIPELINE



- ▶ Entrada del pipeline
 - ▶ El ojo/la cámara:
 - ▶ Posición: matriz "view"
 - ▶ Otras propiedades (perspectiva): matriz "projection"
 - ▶ Por cada luz:
 - ▶ Posición y propiedades (color, dirección, atenuación, etc)

LA RASTERIZACIÓN EN EL PIPELINE



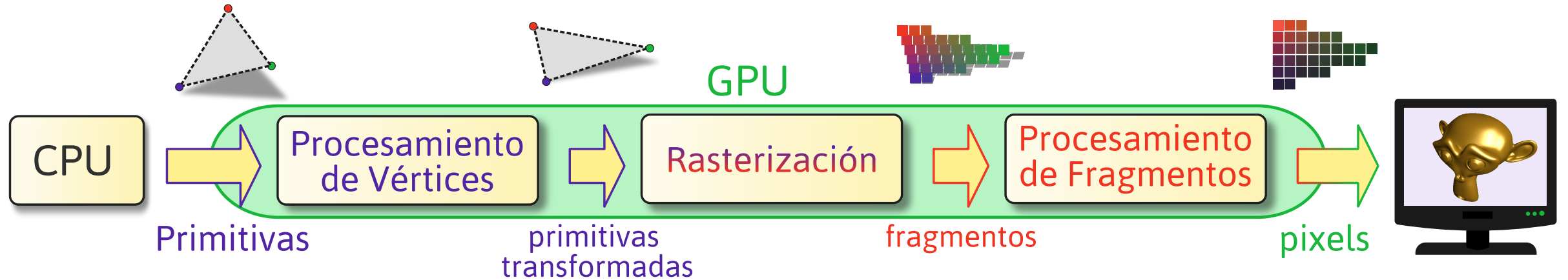
- Etapa vectorial

- Transformación de los vértices (vertex-shader)

! *Unidad "Espacios y Transformaciones"*

- ~~Geometry/Tessellation shaders~~
 - Clipping/culling de primitivas
 - Screen mapping

LA RASTERIZACIÓN EN EL PIPELINE



- La rasterización

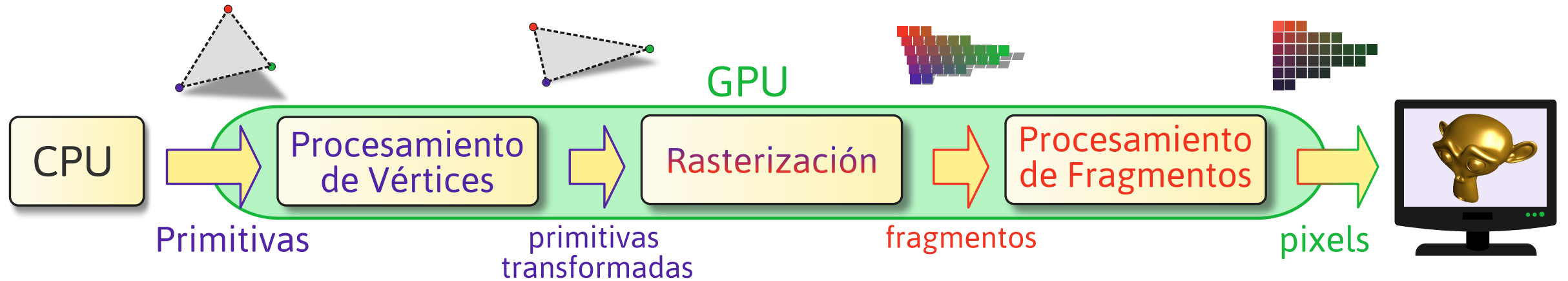
- Por cada triángulo genera fragmentos

! *Unidad "Rasterización"*

- A cada fragmento le "interpola" información adicional

! *Unidad "Interpolación"*

LA RASTERIZACIÓN EN EL PIPELINE



- Etapa Raster

- Iluminación y textura (fragment-shader)

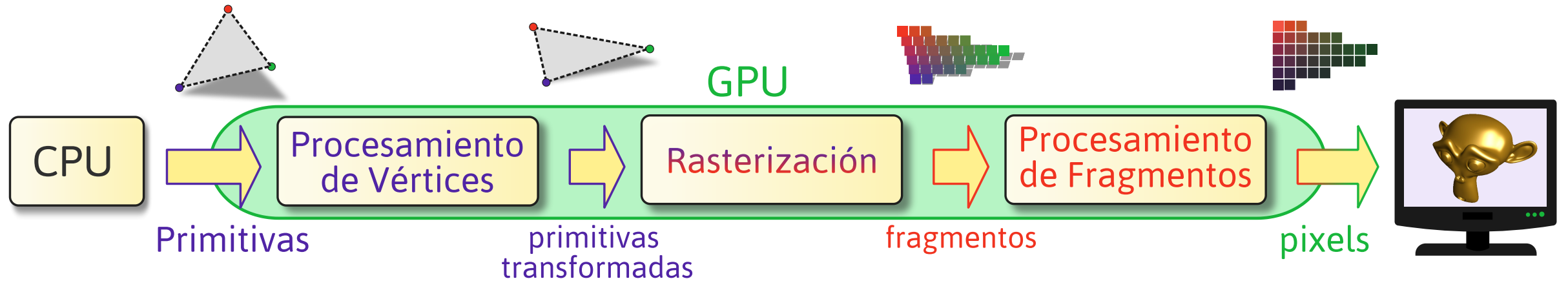
- ! Unidades "Color e Iluminación" y "Texturas"

- Tests (z-buffer, stencil, alpha)

- ! Unidad "Técnicas del espacio de la imagen"

- Blending

LA RASTERIZACIÓN EN EL PIPELINE



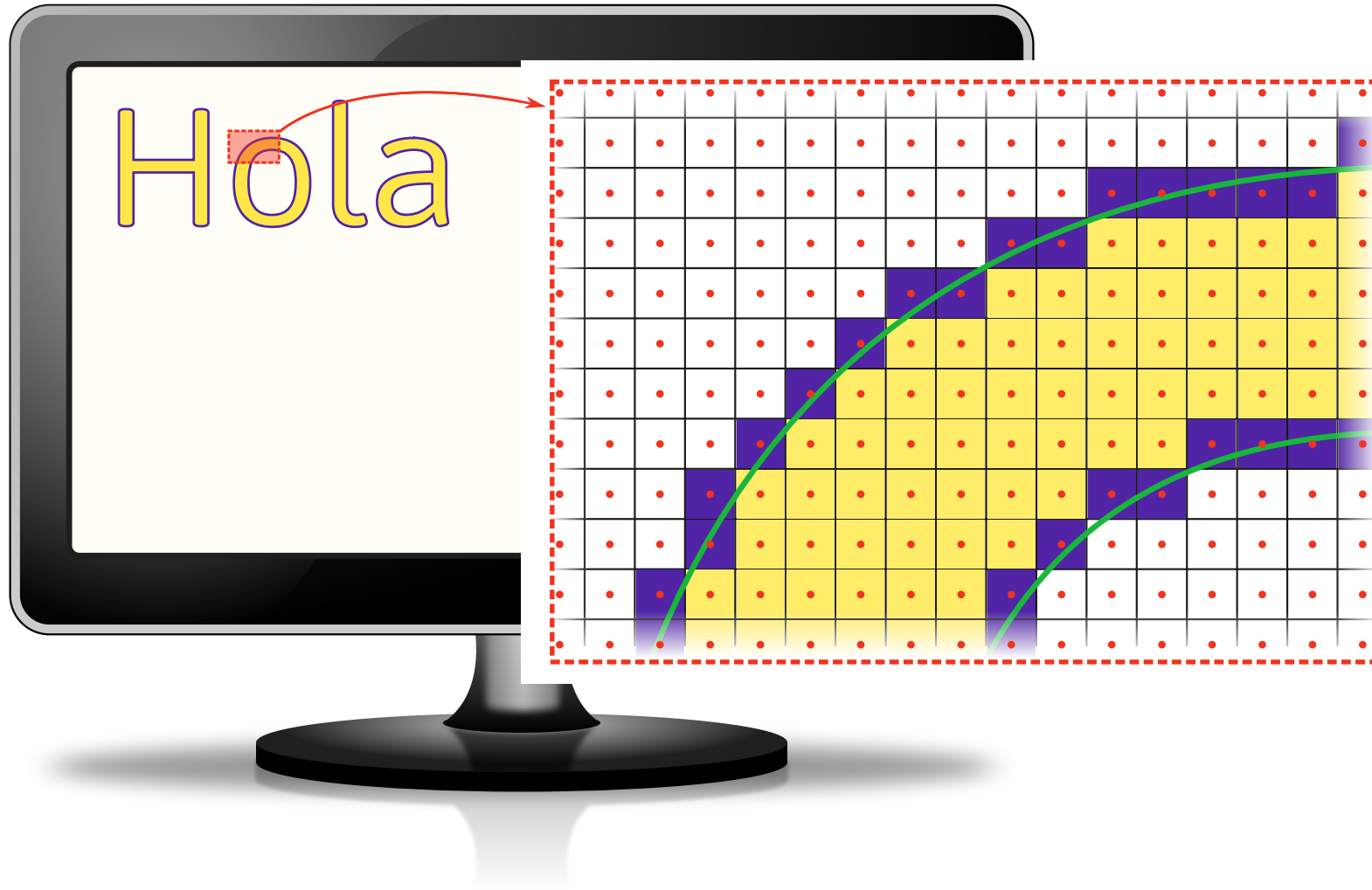
- Salida del pipeline
 - La imagen raster para mostrar en pantalla
 - Algunas otras imagenes auxiliares (buffers)

Raster

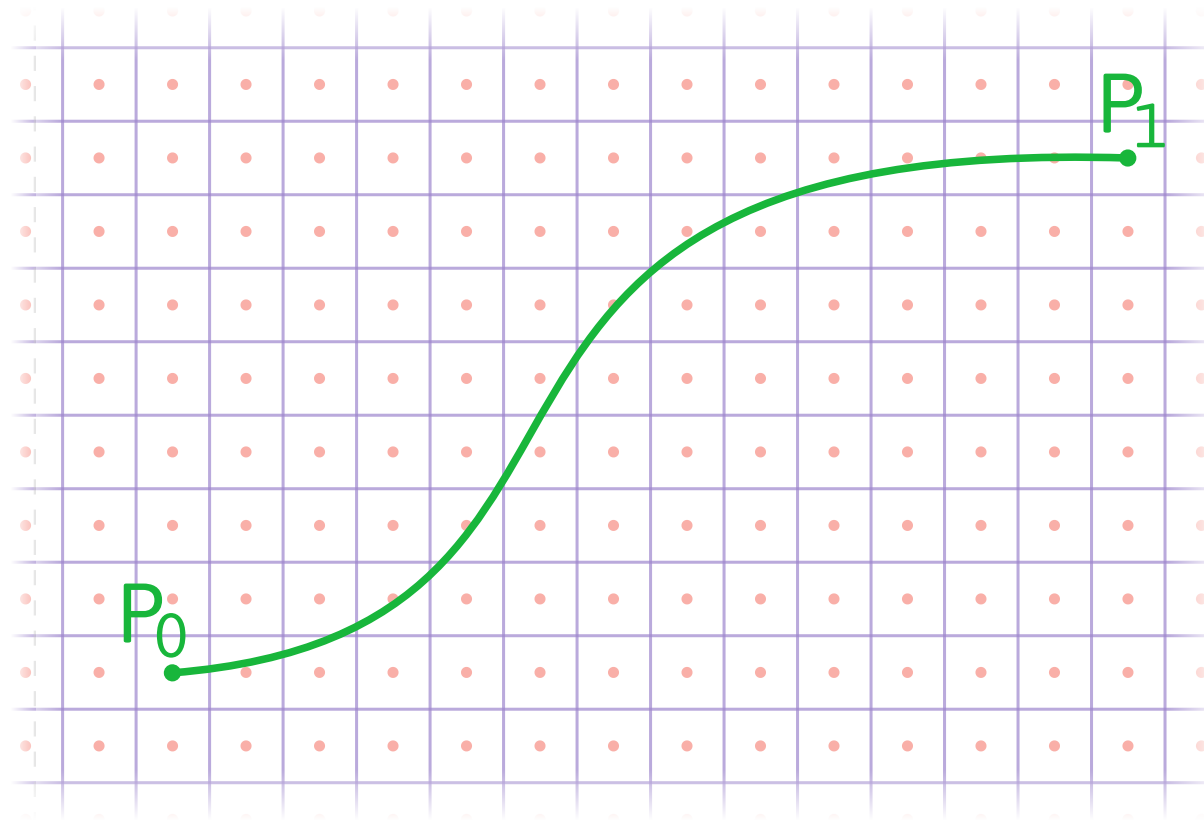
VS

Vector

RASTERIZACIÓN



REQUERIMIENTOS



1. Representación Adecuada
2. **Contigüidad** (water-tight)
3. Eficiencia y Robustez

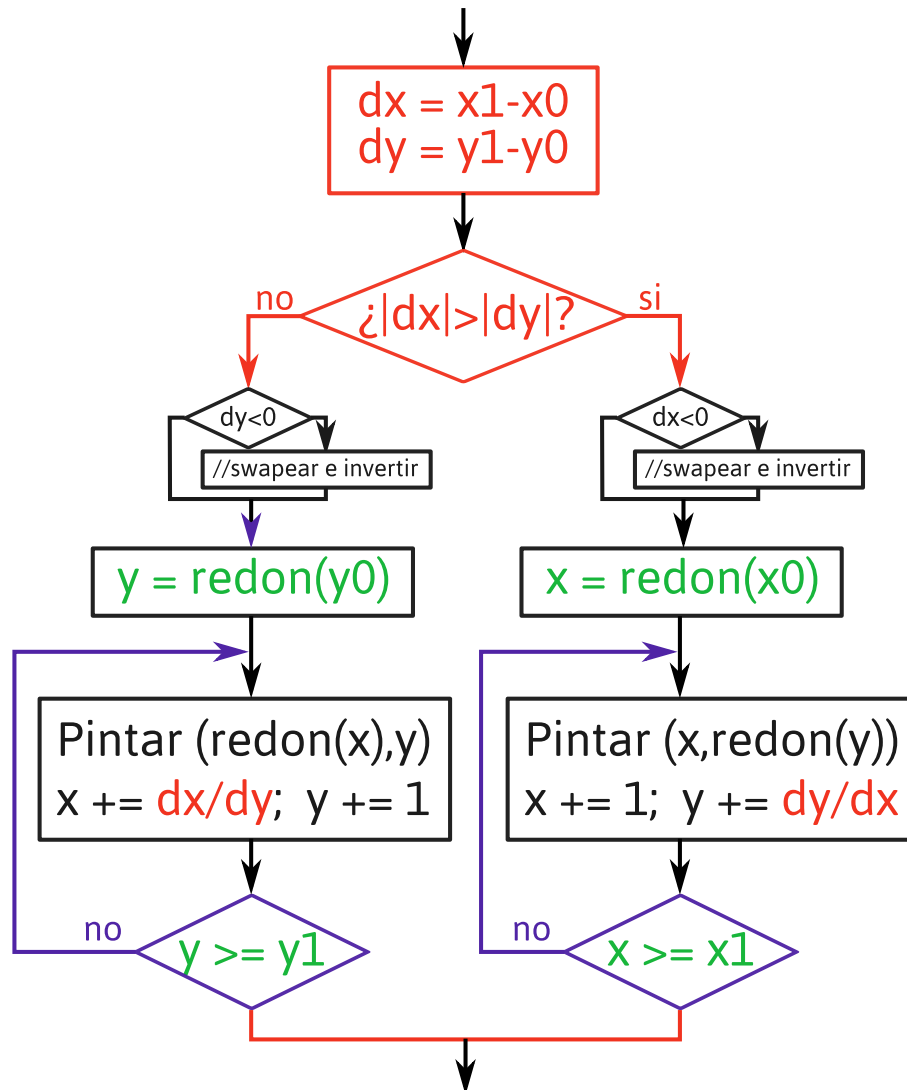
$$\max(\Delta x, \Delta y) \leq 1$$

A 2D grid with a light purple background and a grid of thin purple lines. Red dots are placed at the intersections of the grid lines. A green line segment connects two points, labeled P_0 and P_1 . P_0 is located at the intersection of the 2nd vertical line from the left and the 8th horizontal line from the bottom. P_1 is located at the intersection of the 17th vertical line from the left and the 13th horizontal line from the bottom. The line segment represents a path from P_0 to P_1 .

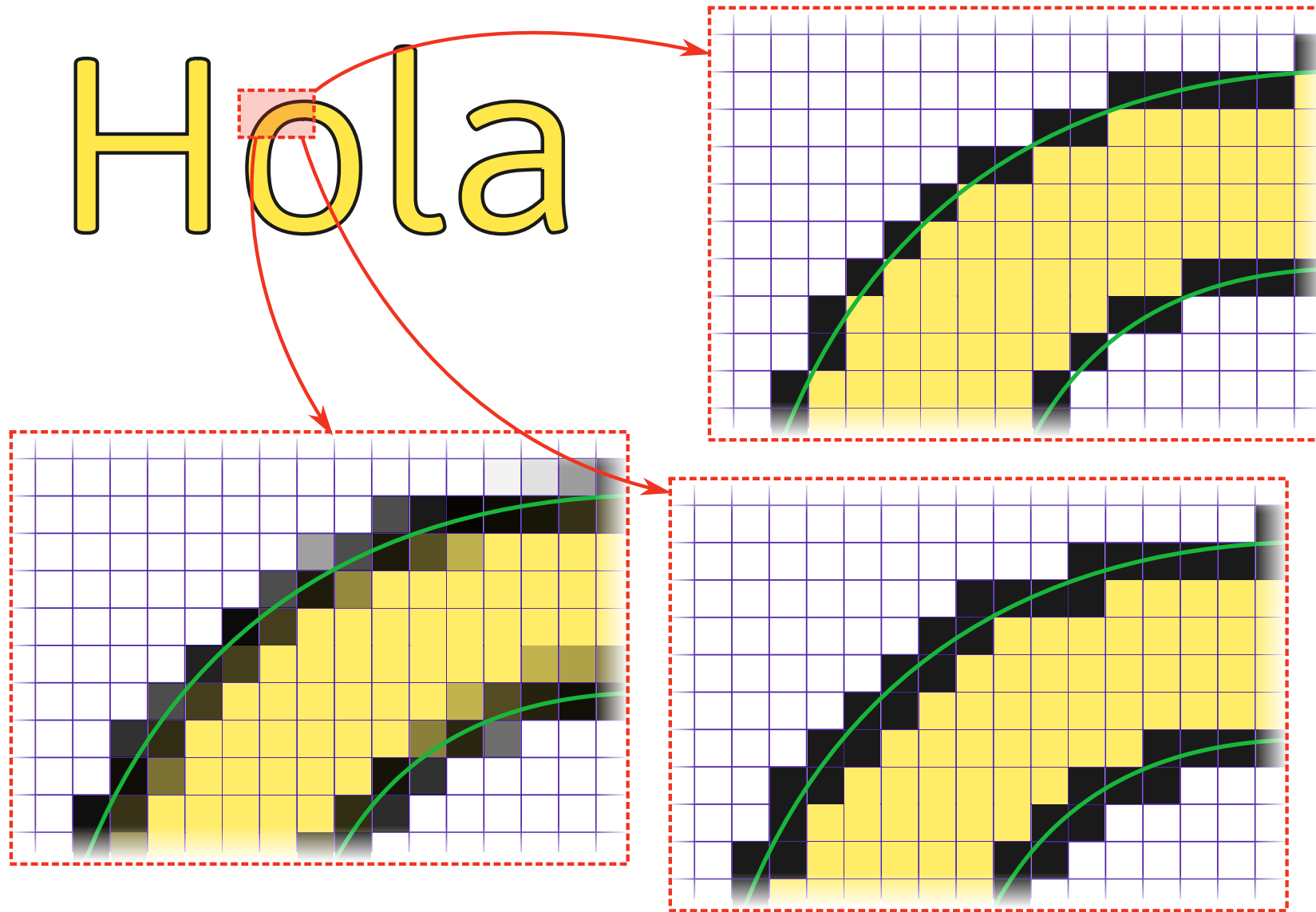
LINEA DDA

Linea DDA

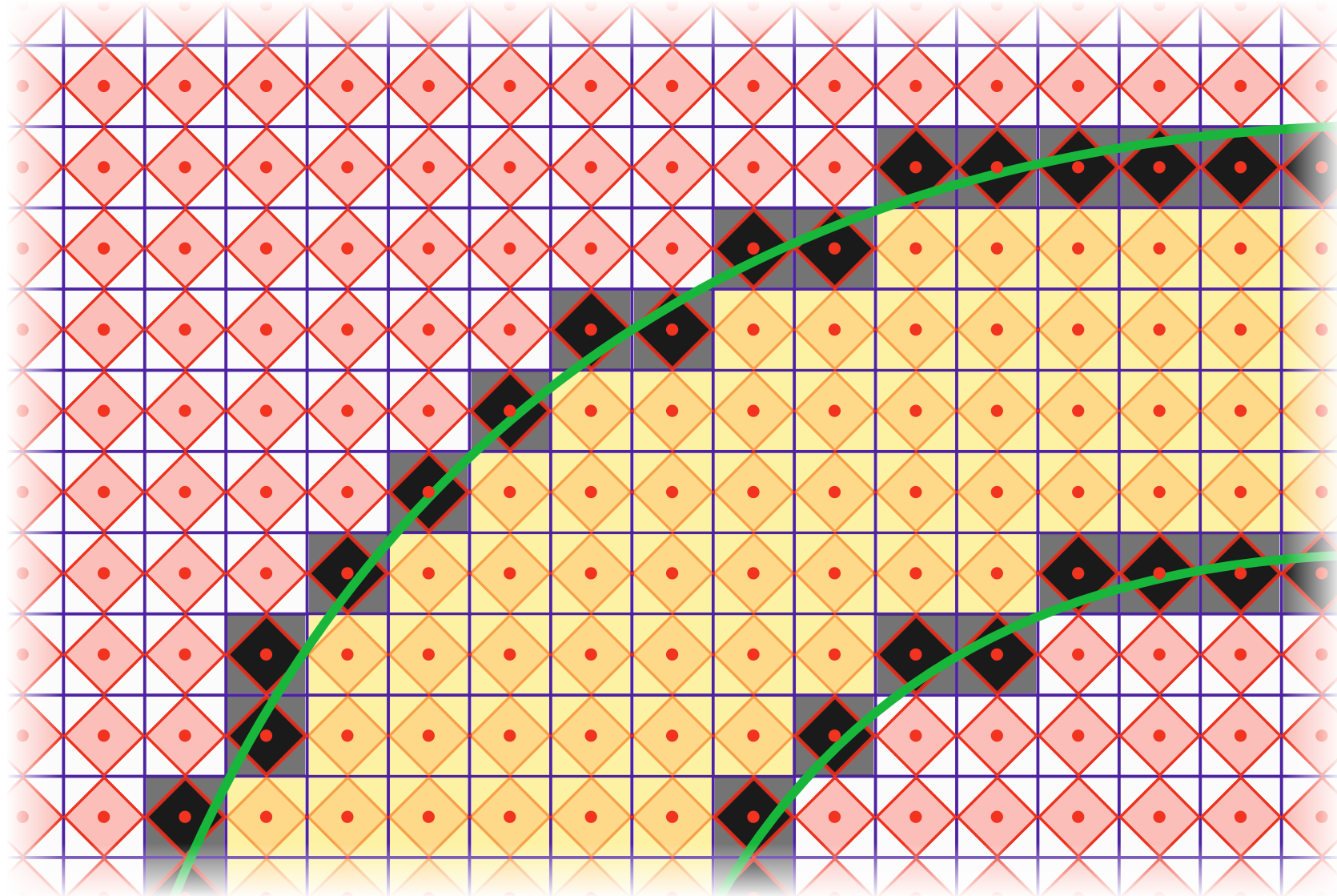
in: x_0, y_0, x_1, y_1



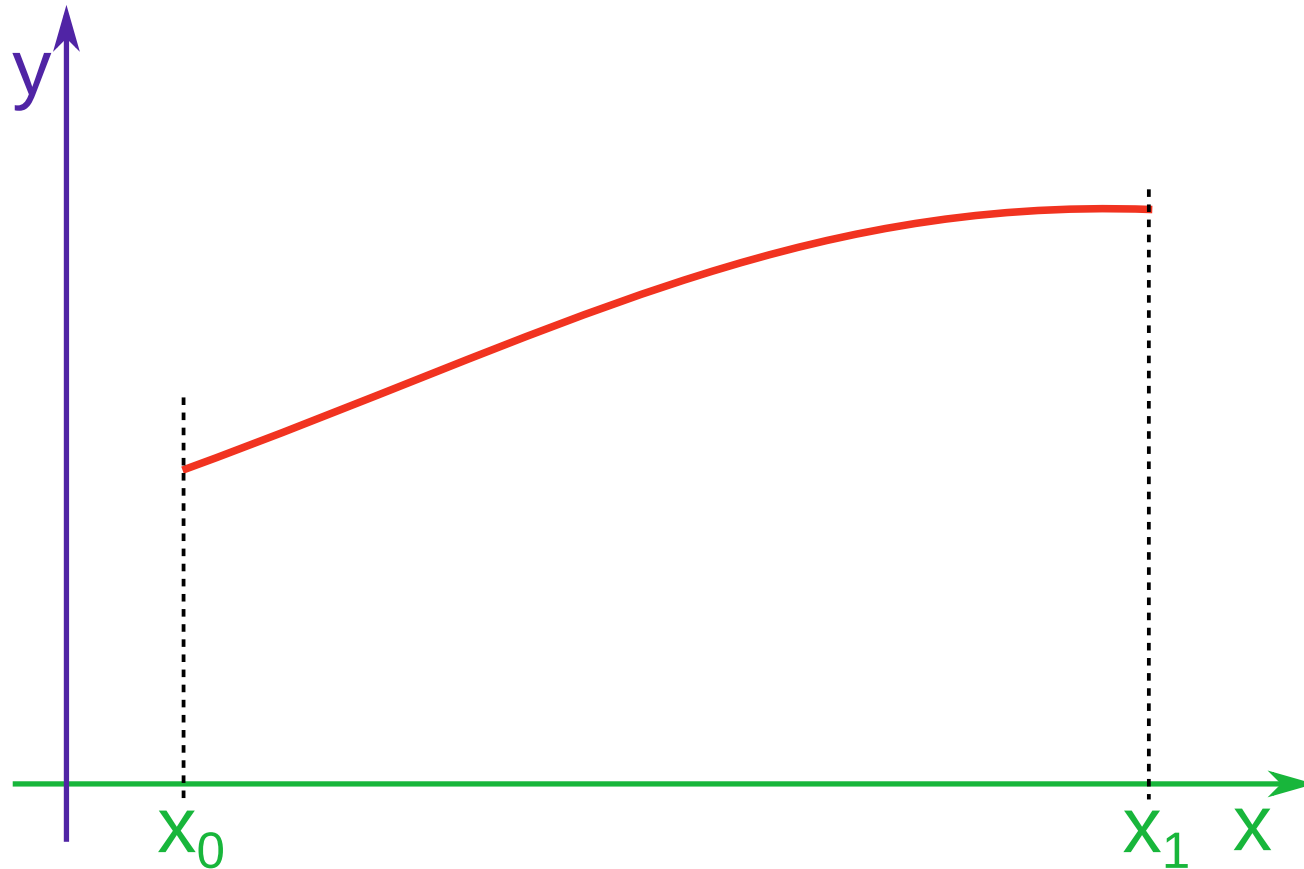
RASTERIZACIÓN



RASTERIZACIÓN



CURVAS PARAMÉTRICAS

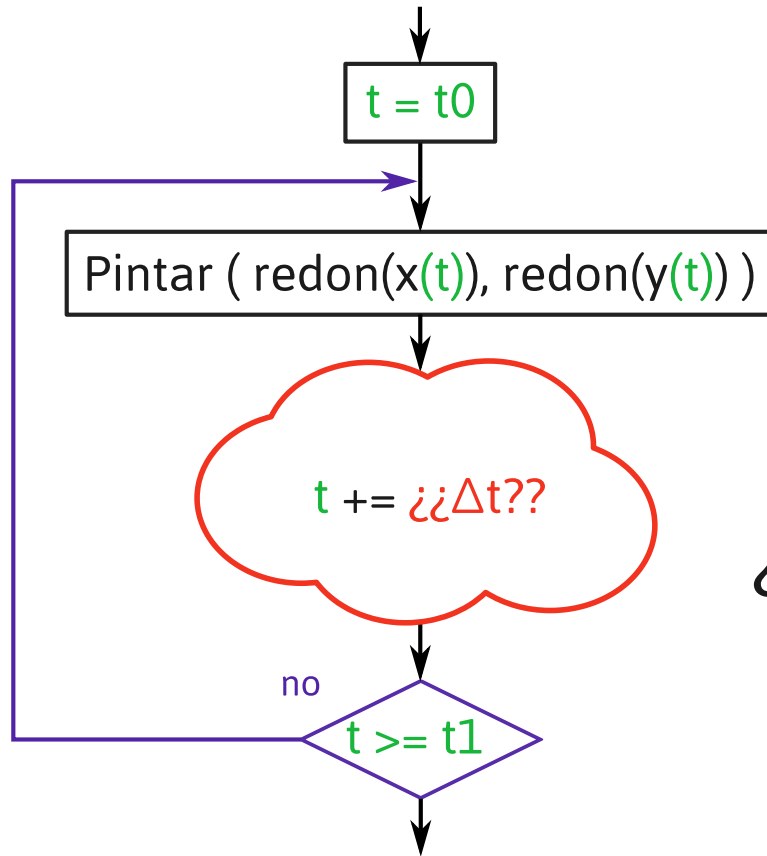


$$y = f(x) \quad x \in [x_0; x_1]$$

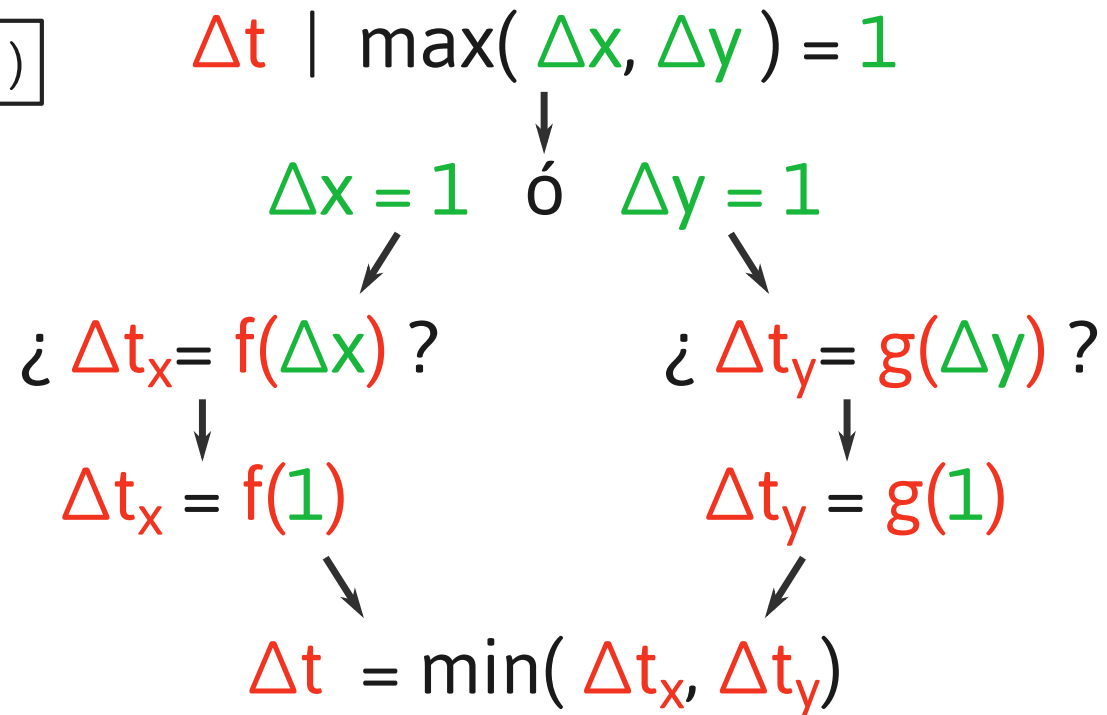
RASTERIZACIÓN DE CURVAS POR DERIVADAS

Curva DDA

in: $x(t)$, $y(t)$, $dx(y)$, $dy(t)$, t_0 , t_1

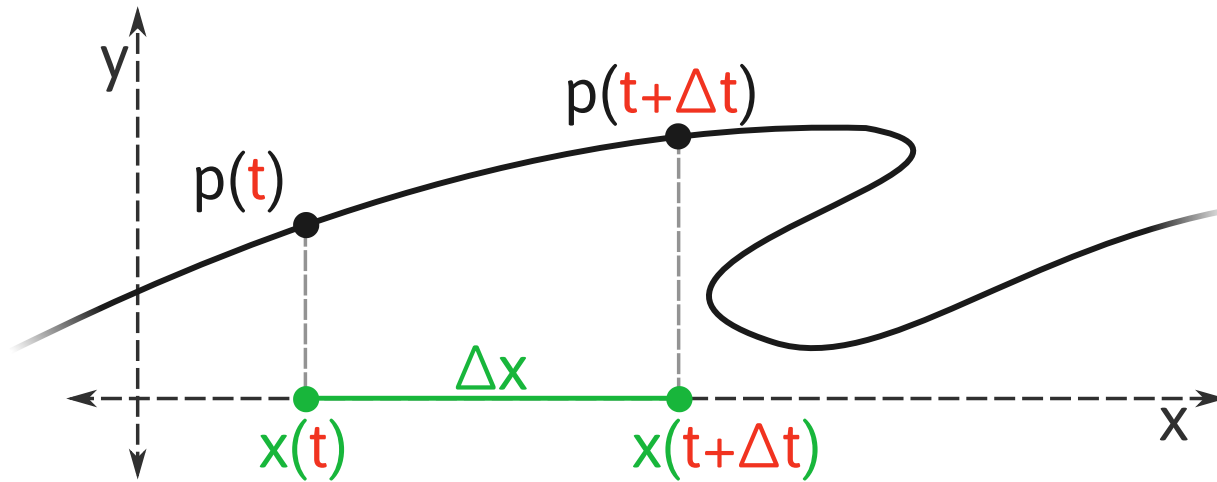


contigüidad: $\max(\Delta x, \Delta y) \leq 1$



RASTERIZACIÓN DE CURVAS POR DERIVADAS

¿ Δt | $\Delta x = 1$?



$$\Delta x = x(t+\Delta t) - x(t)$$

$$\leftarrow x(t+\Delta t) \approx x(t) + \Delta t x'(t)$$

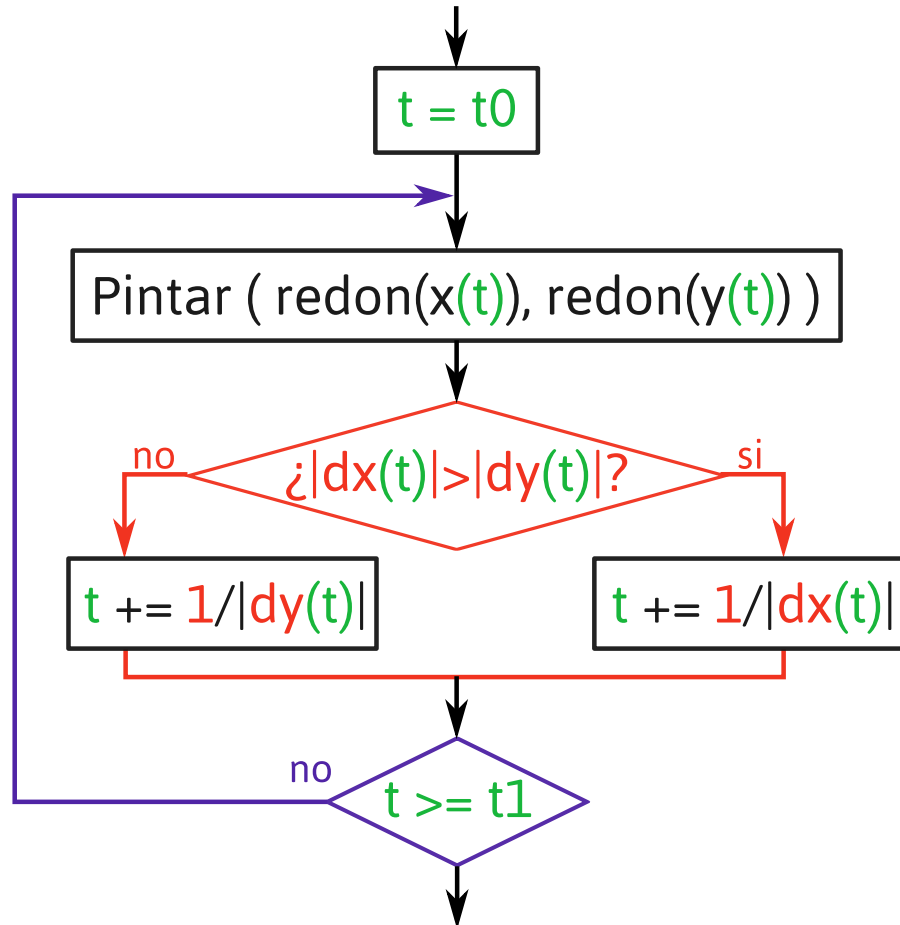
$$\Delta x \approx x(t) + \Delta t x'(t) - x(t)$$

$$\Delta x \approx \Delta t x'(t) \rightarrow 1 \approx \Delta t x'(t) \rightarrow \Delta t \approx 1 / x'(t)$$

RASTERIZACIÓN DE CURVAS POR DERIVADAS

Curva DDA

in: $x(t), y(t), dx(t), dy(t), t_0, t_1$

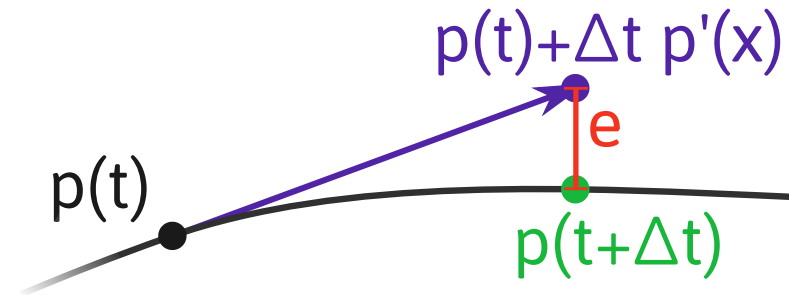


$$\Delta t = \min(\Delta t_x, \Delta t_y)$$

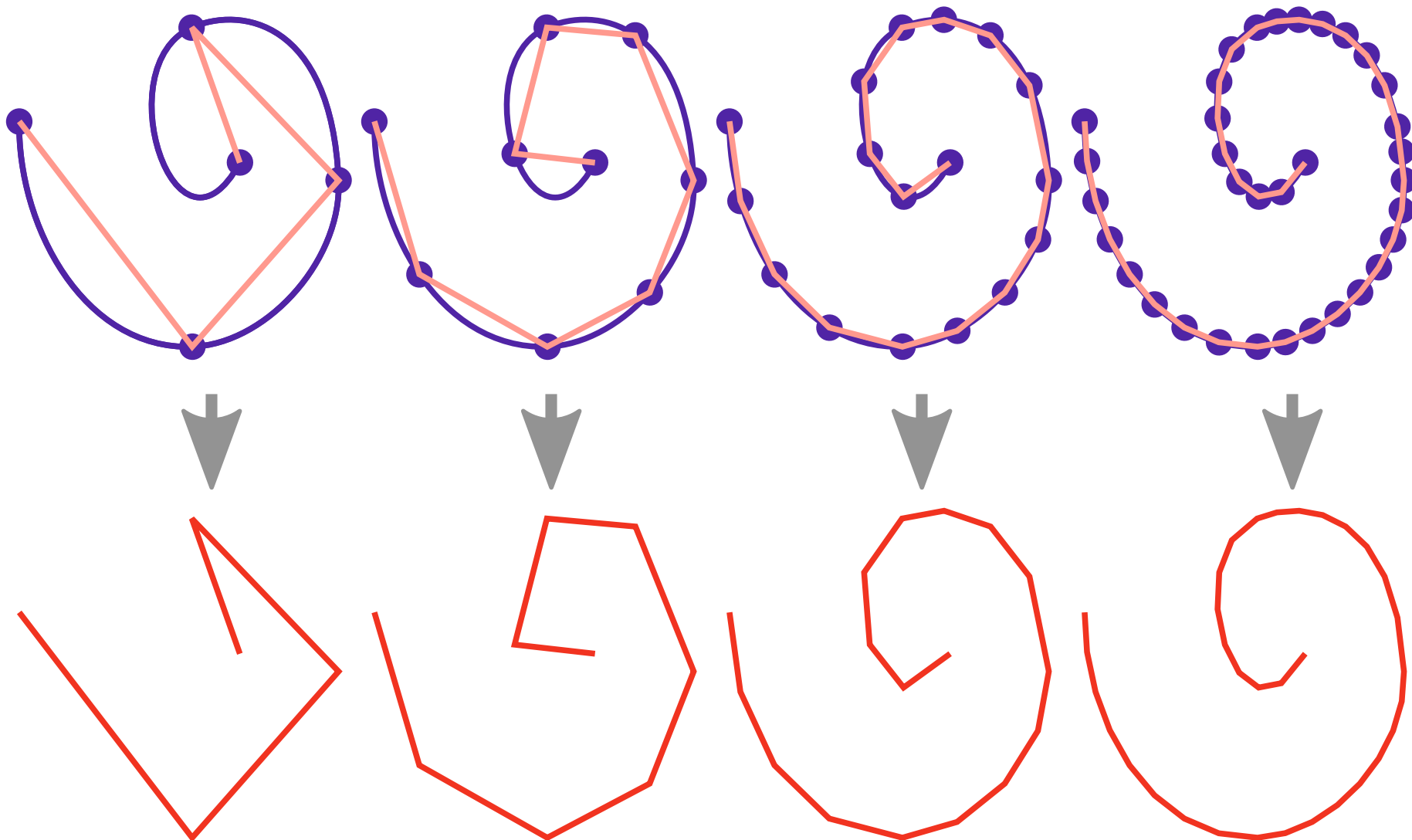
$$\Delta t_x \approx 1 / x'(t)$$

$$\Delta t = \min(1/|x'(t)|, 1/|y'(t)|)$$

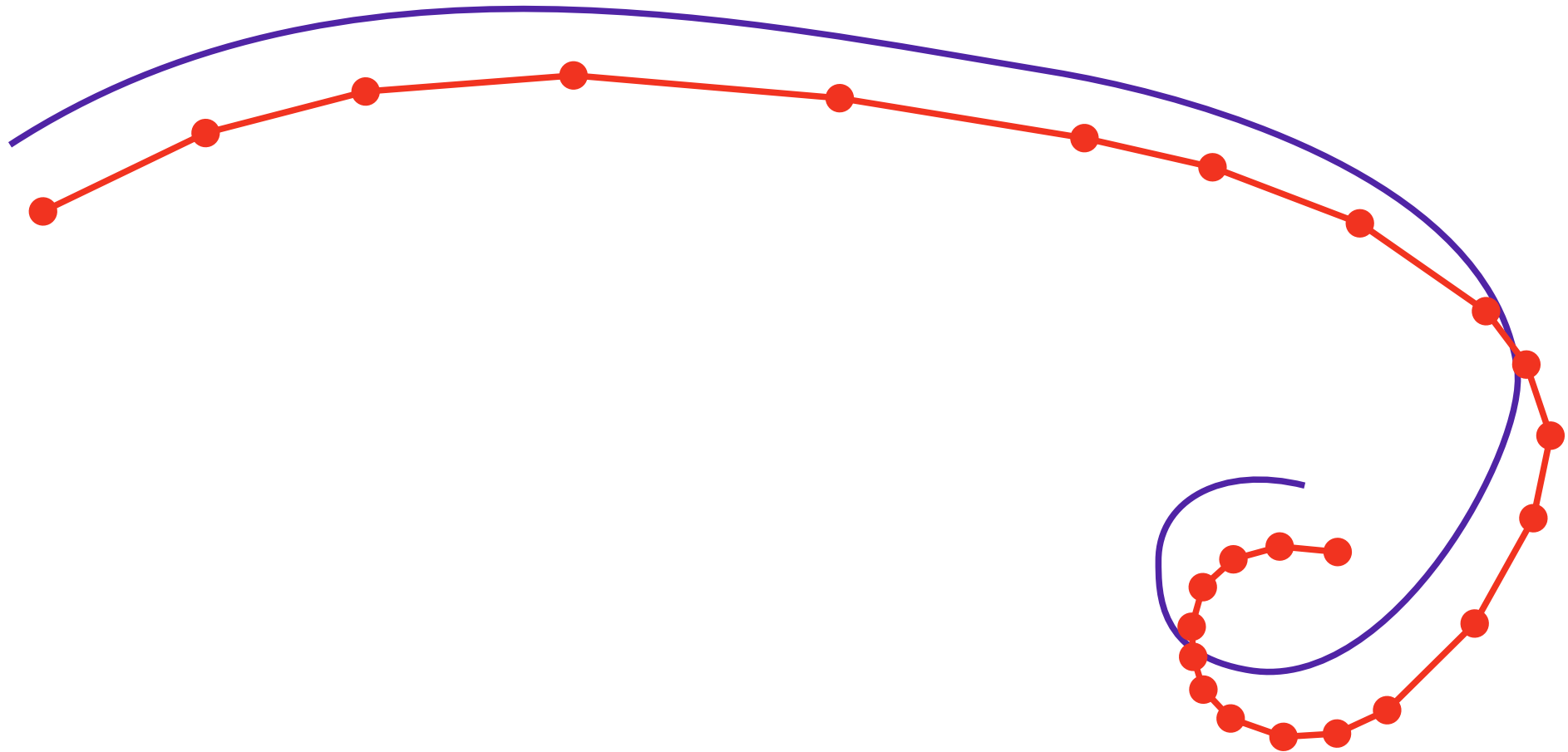
$$\Delta t = 1 / \max(|x'(t)|, |y'(t)|)$$



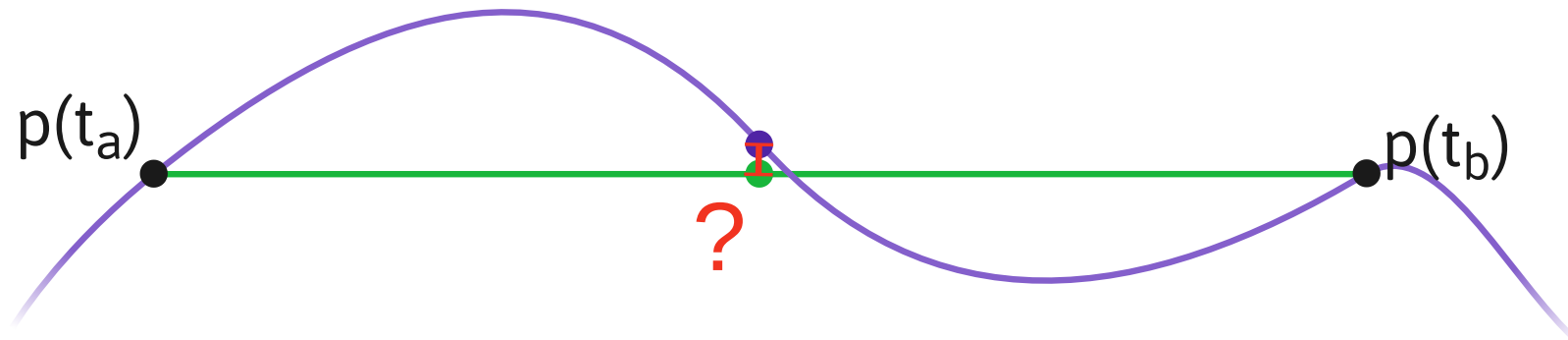
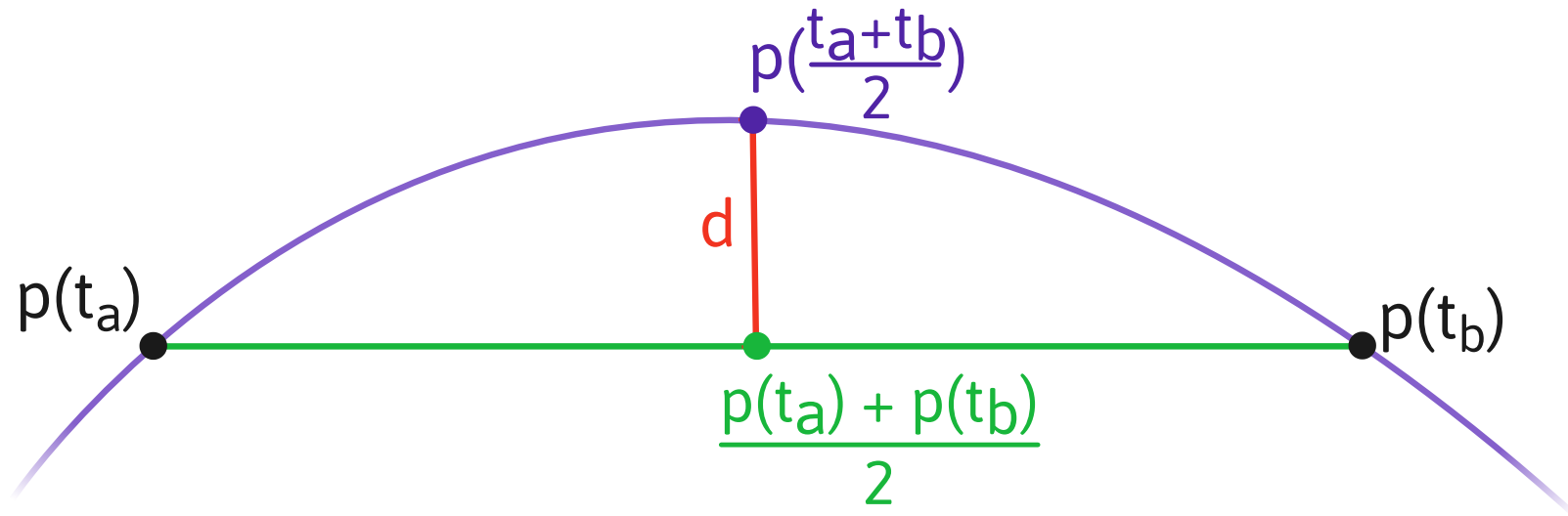
RASTERIZACIÓN POR SUBDIVISIONES



RASTERIZACIÓN POR SUBDIVISIONES



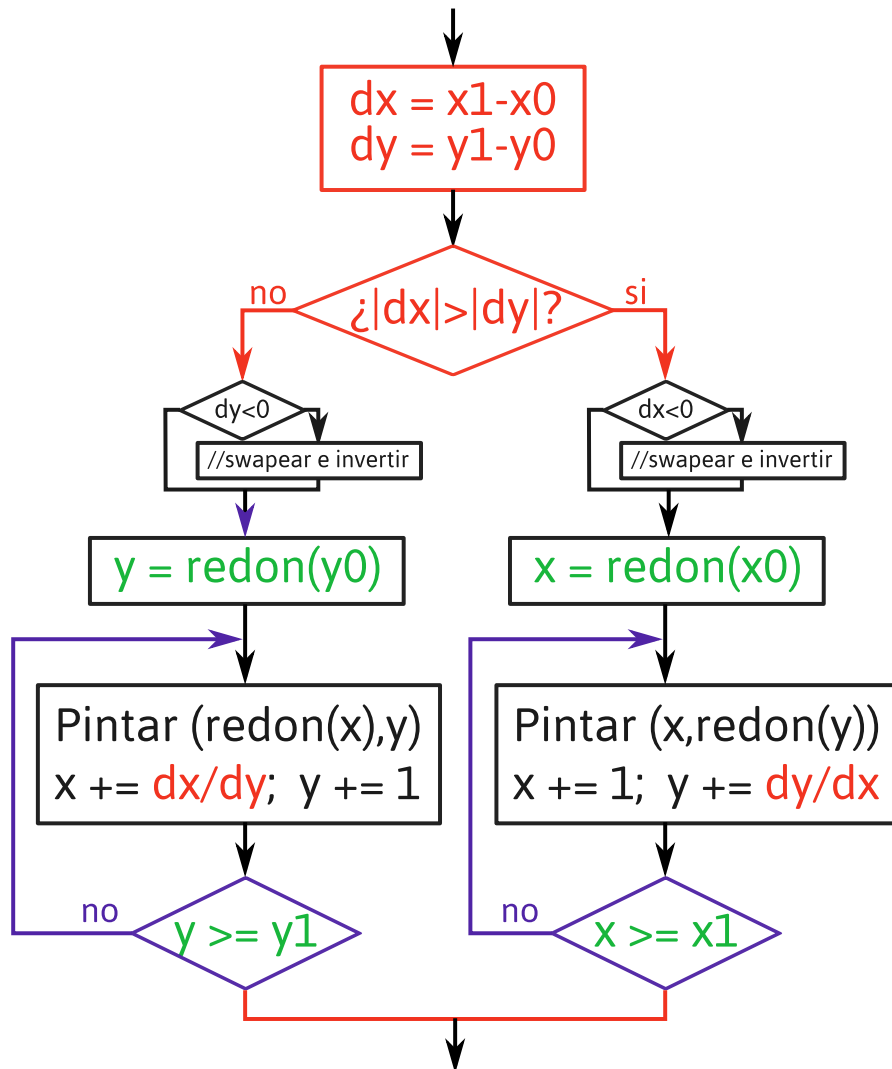
RASTERIZACIÓN POR SUBDIVISIONES



RASTERIZACIÓN POR DDA: LINEA VS CURVA

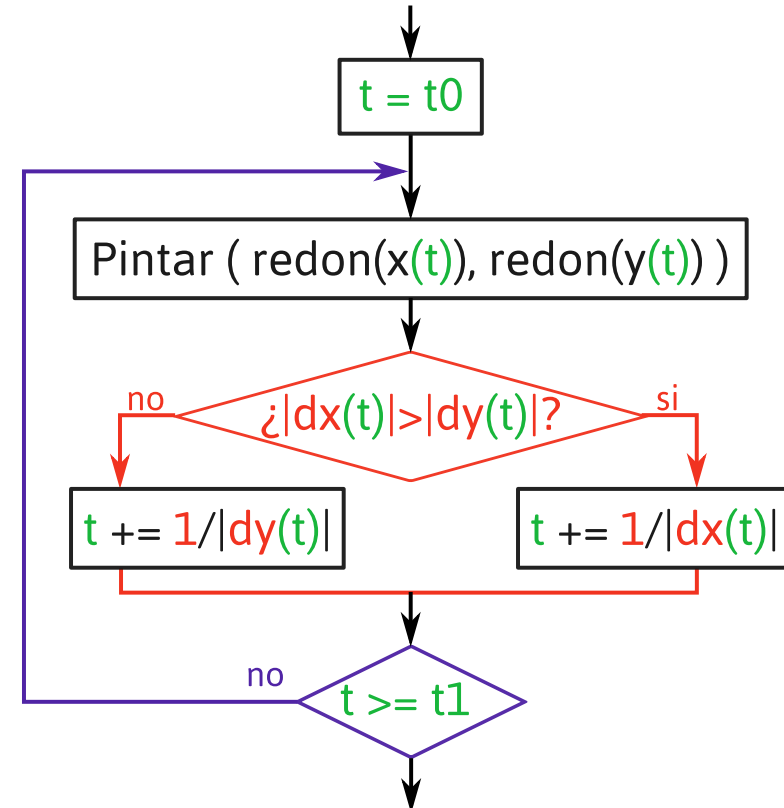
Linea DDA

in: x_0, y_0, x_1, y_1

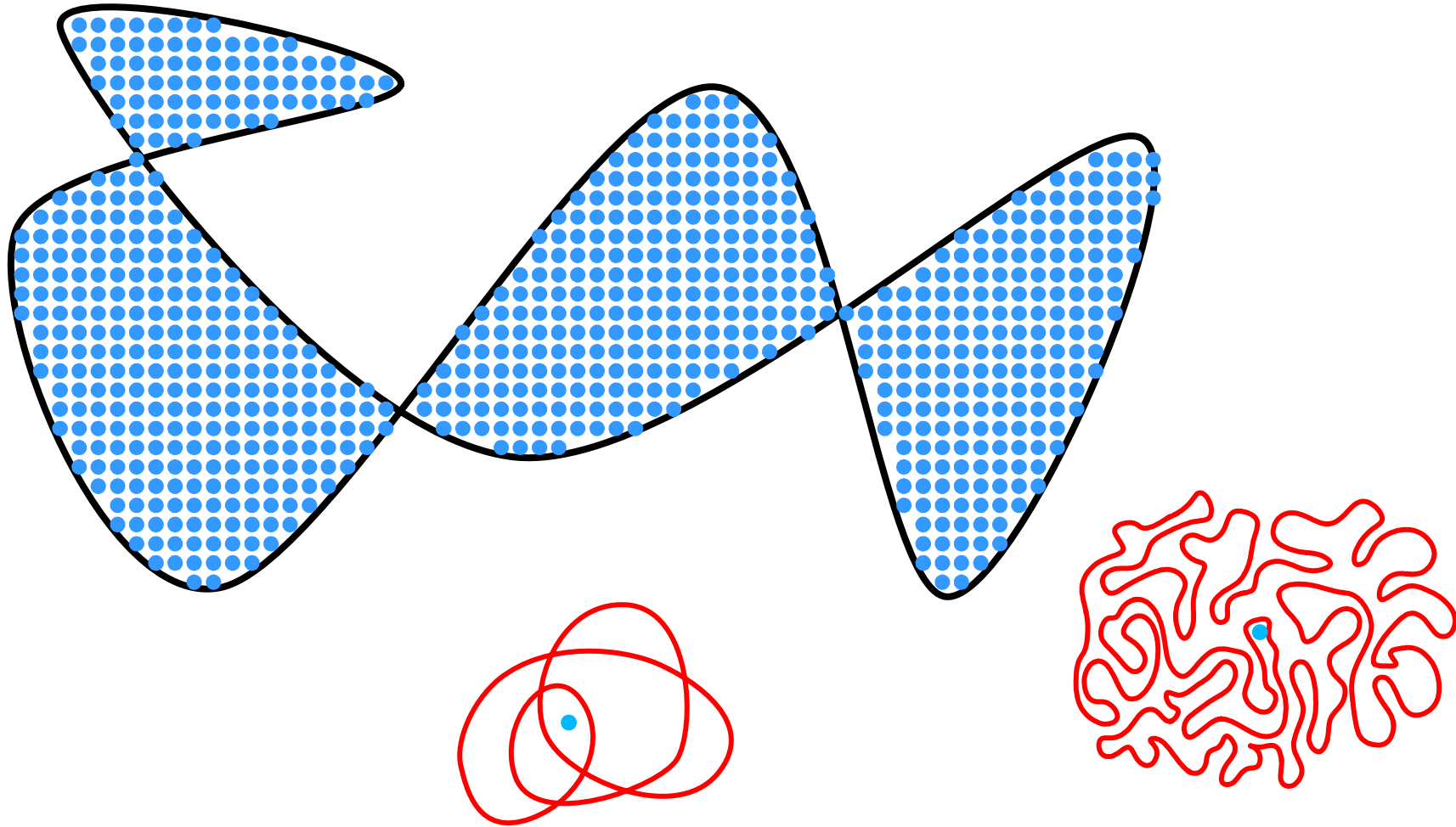


Curva DDA

in: $x(t), y(t), dx(t), dy(t), t_0, t_1$

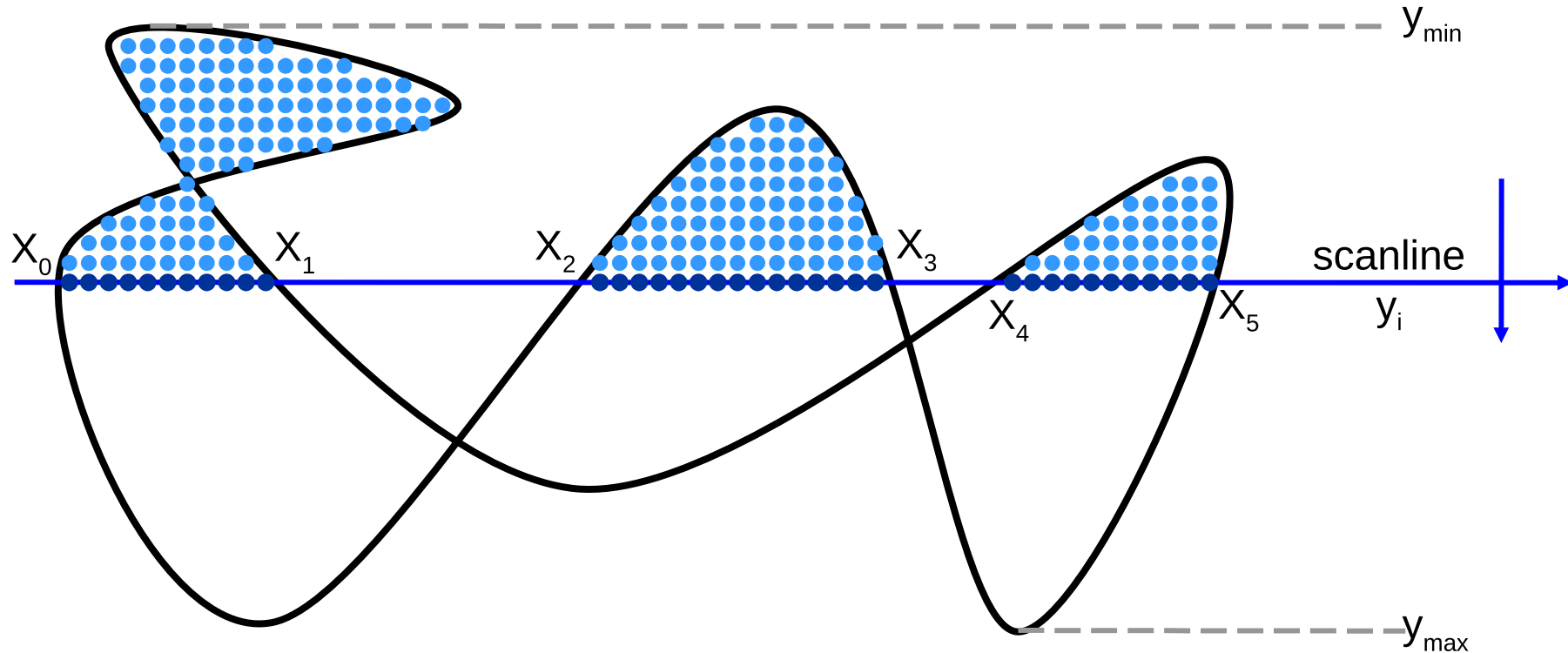


RASTERIZACIÓN DE FIGURAS CERRADAS



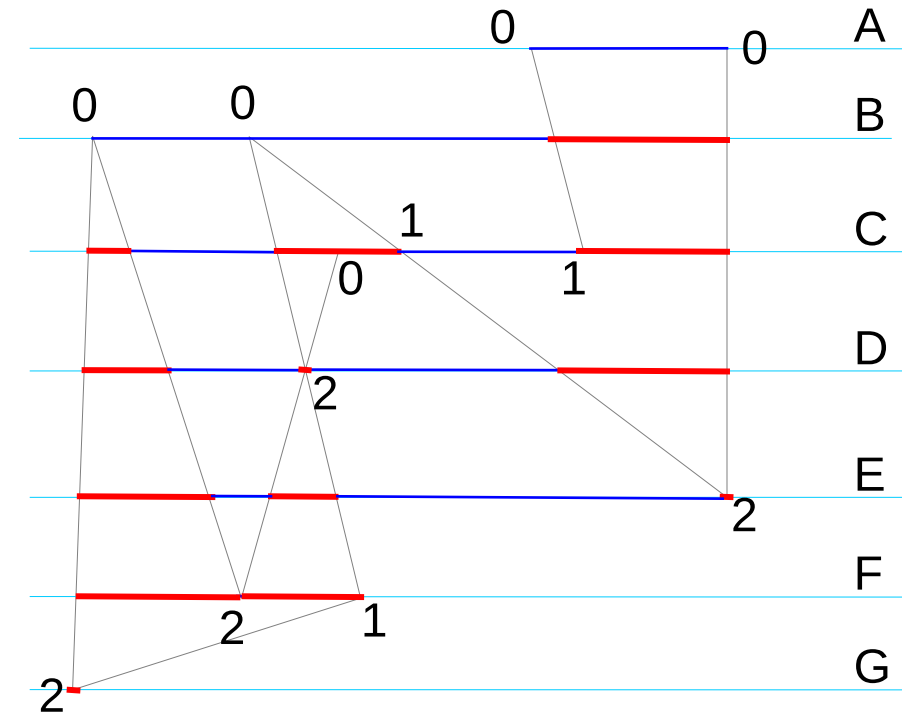
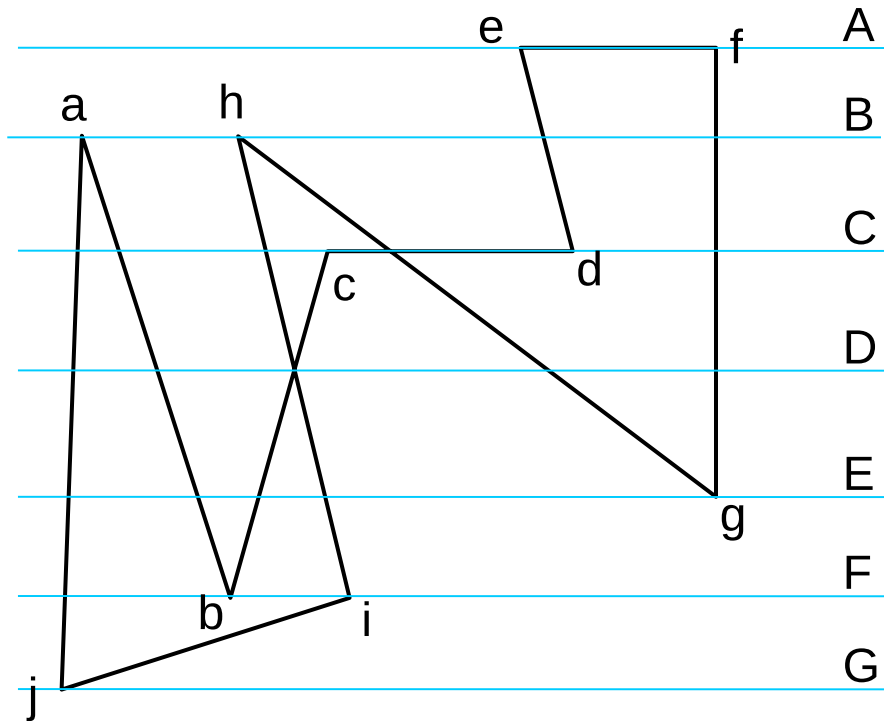
RASTERIZACIÓN DE FIGURAS CERRADAS

Igual que el *buffer* de color *cache friendly*



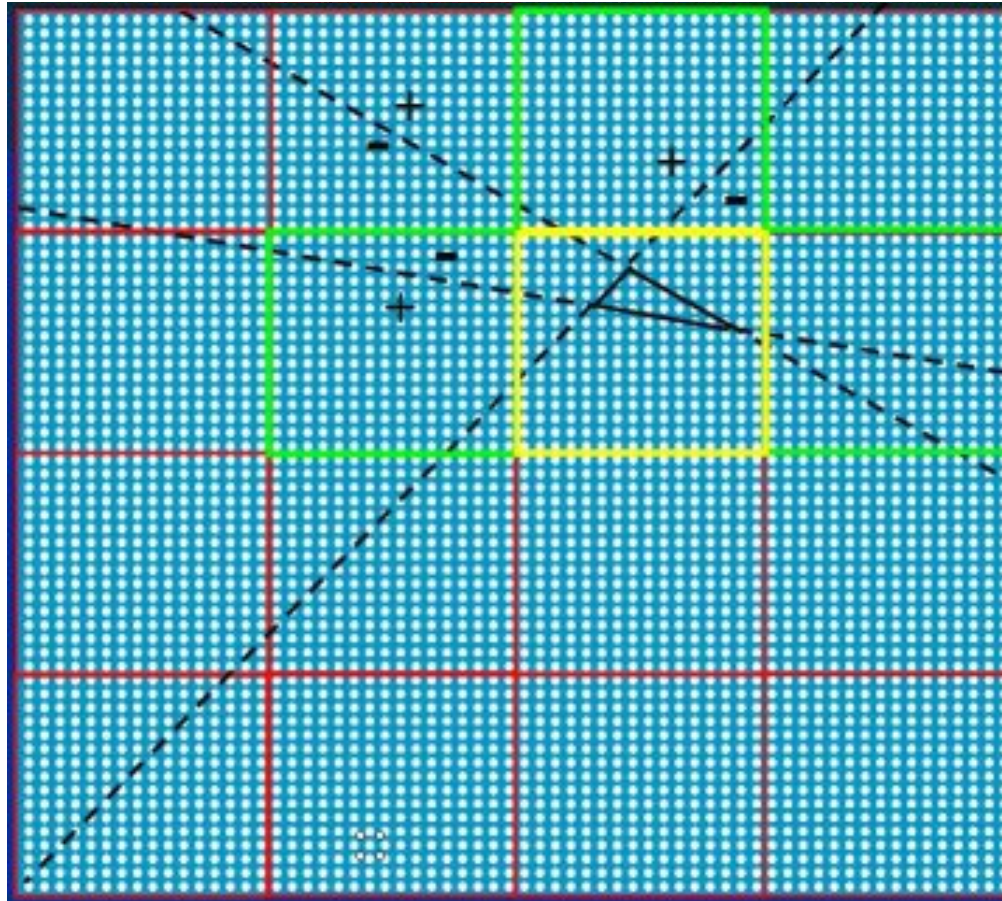
```
y=ymin+1; i=1; while(y<ymax) {  
    j=0; while(j<n[i]-1) {  
        x=X[i][j]+1; while(x<X[i][j+1]){pinta(x,y); x++;} j+=2;  
    }  
    y++; i++;  
}
```

RASTERIZACIÓN DE POLIGONALES

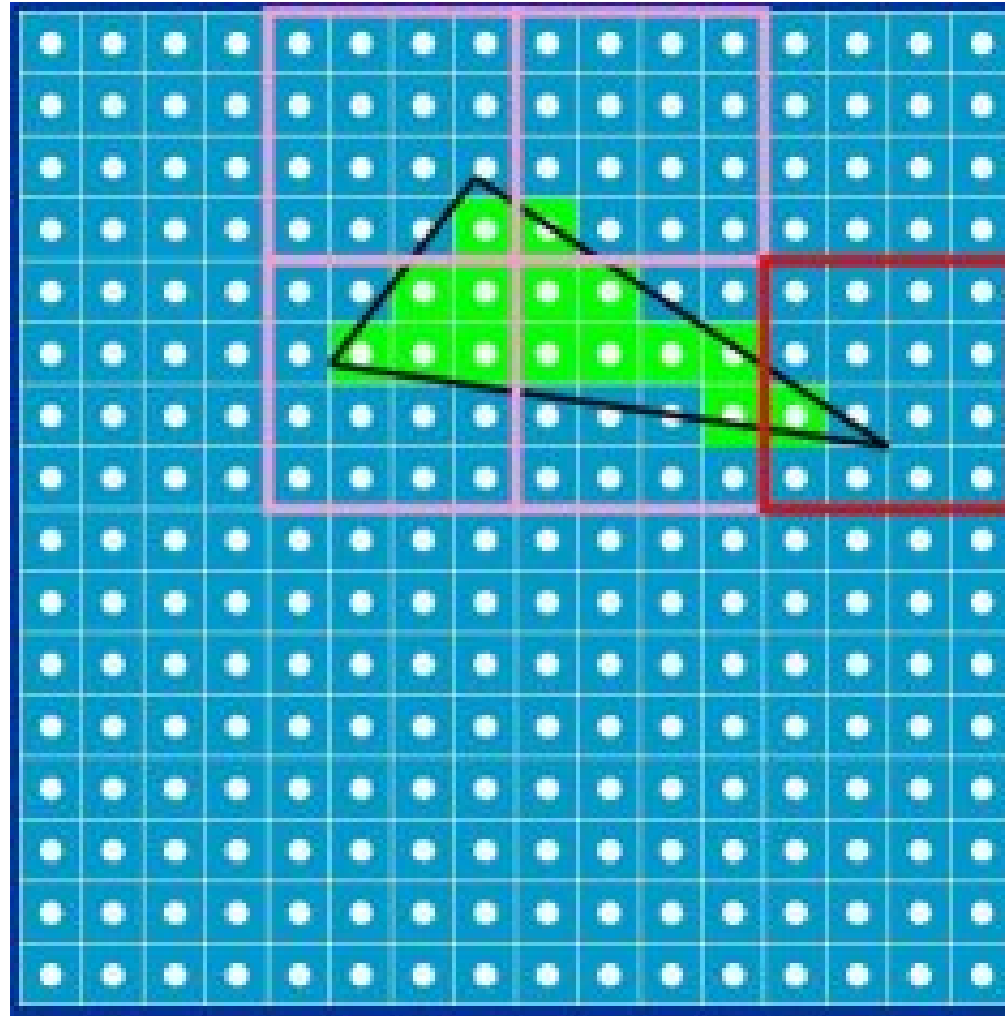


Pasos horizontales no se agregan a la lista.
Vértices: sólo los mínimos estrictos de su segmento.

RASTERIZACIÓN DE TRIÁNGULOS



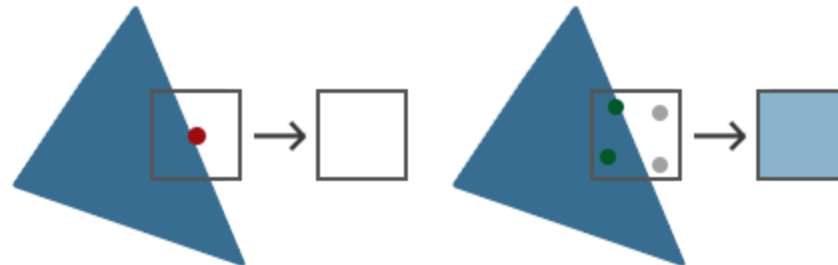
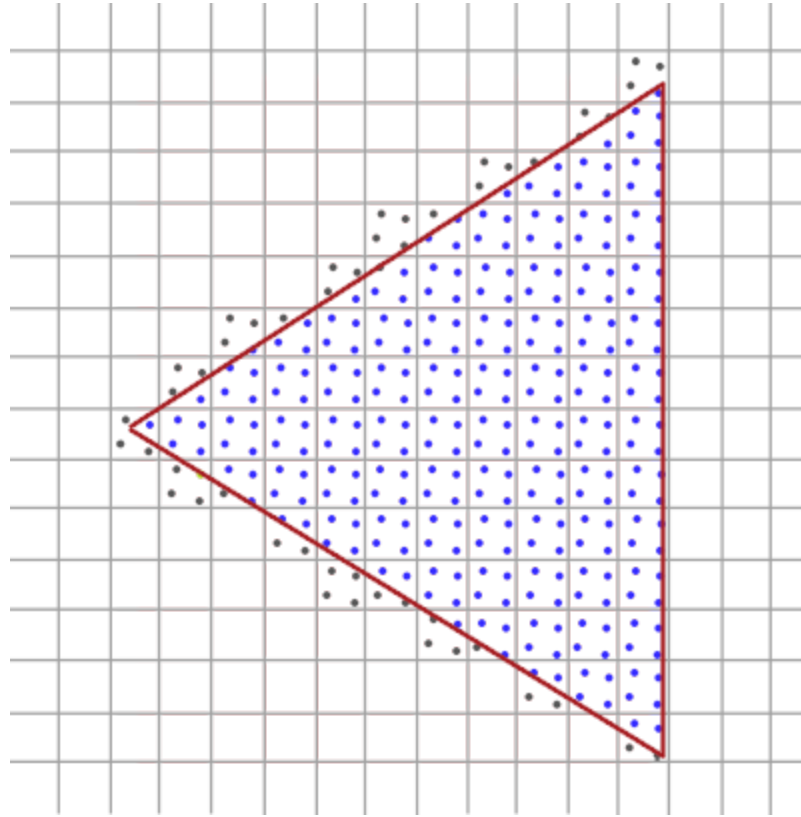
RASTERIZACIÓN DE TRIÁNGULOS



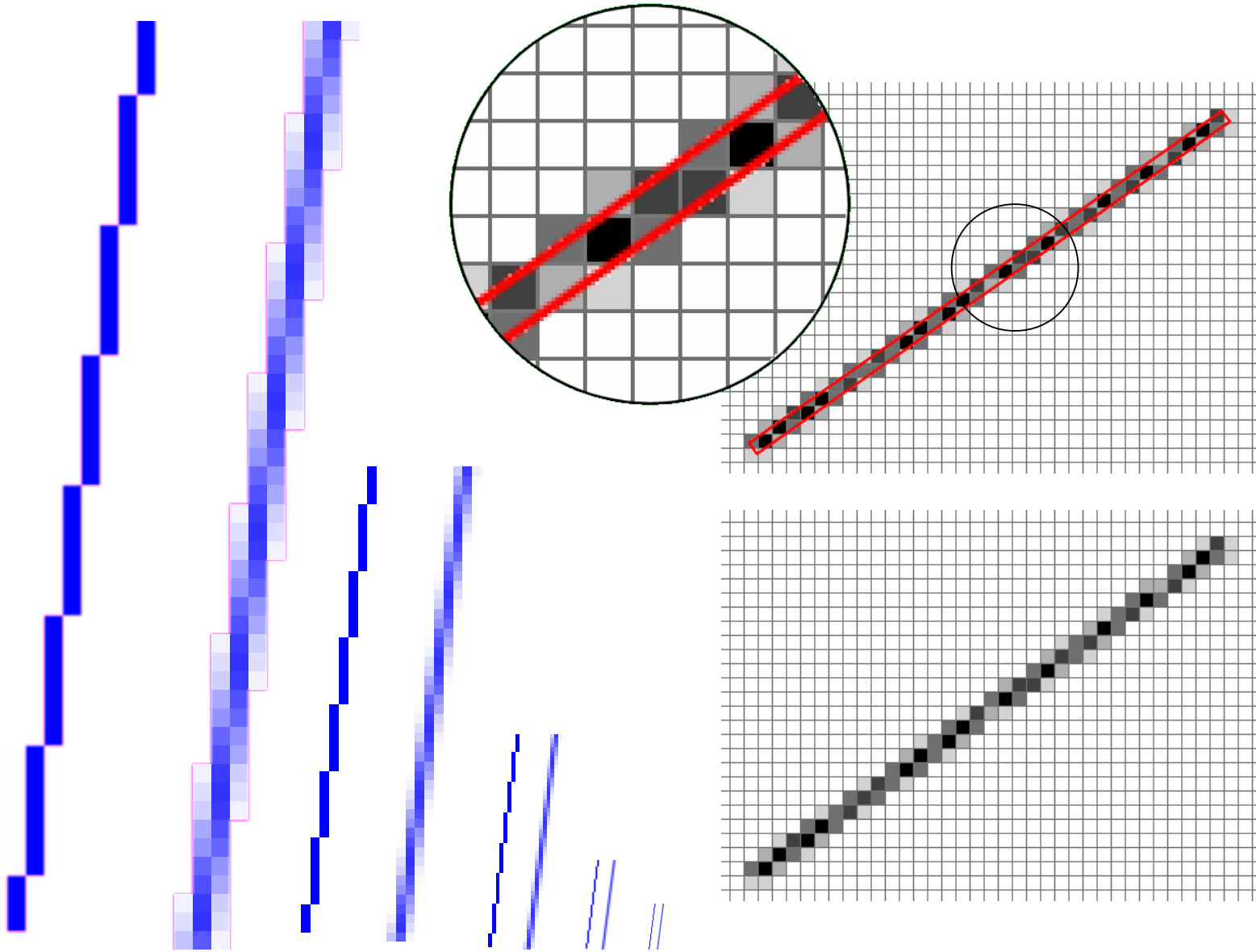
ANTI-ALIASING



ANTI-ALIASING



ANTI-ALIASING



¿QUÉ HAY QUE SABER?

- ▶ [3] Etapas generales del pipeline: (ídem introducción)
 - ▶ Objetivo y entrada/salida de cada una
- ▶ Vector vs. Raster:
 - ▶ En qué consiste, ventajas y desventajas
- ▶ Rasterización:
 - ▶ Definición del problema y requerimientos
 - ▶ Algoritmos DDA (segmentos y curvas), y subdivisión (curvas)
 - ▶ Rasterización de figuras: solo test del rayo e idea de scanline
 - ▶ Rasterización de triángulos: solo lineamientos generales
 - ▶ Antialiasing: solo ¿qué es? ~~(tipos/algoritmos no)~~